Chapter Contents

Chapter 13 - Initial-Value Problems

Chapters 9 through 12 developed methods for finding general solutions of linear differential equations. A general solution contains arbitrary constants because the differential equation alone describes an entire family of functions.

An **initial-value problem**, or **IVP**, adds enough information at one point to select one member of that family.

This chapter focuses on the stage that is often compressed into the phrase "apply the initial conditions." That phrase hides several decisions:

- which solution must receive the conditions?
- how many derivatives must be calculated?
- how are the resulting constant equations formed?
- why must those equations usually be solved simultaneously?
- what changes when the initial point is not zero?
- why do zero initial data imply a zero solution in one problem but not in another?
- how can the Wronskian explain the uniqueness of the selected constants?
- how are higher-order initial-value problems handled?

The guiding rule is:

> **solve the differential equation first, write the complete general solution, and only then use all initial conditions to determine its arbitrary constants.**

All examples, equations, and exercises in this chapter are independently constructed. The reference text is used only to identify the mathematical topics.

What An Initial-Value Problem Adds

The Goal Of This Block

The goal is to separate the two parts of an initial-value problem:

1. the differential equation determines a family of possible solutions
2. the initial conditions select one member of that family

Neither part should be silently substituted into the other before the general solution has been found.

Equation, Conditions, And Selected Solution

A second-order initial-value problem has the form:

$$a(x)y'' + b(x)y' + c(x)y = g(x),$$

with two conditions at the same point x_0:

$$y(x_0) = Y_0, \qquad y'(x_0) = Y_1.$$

Here:

- x_0 is the initial point
- Y_0 is the prescribed value of the function
- Y_1 is the prescribed value of its first derivative

The differential equation is satisfied throughout an interval. The two initial conditions are satisfied at the single point x_0.

The practical meaning is:

the equation controls how a solution evolves, while the initial data specify where that evolution begins and in which direction it initially moves.

Why A Second-Order Problem Needs Two Conditions

A second-order general solution normally contains two arbitrary constants:

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + y_p(x).$$

To determine both c_1 and c_2, two independent scalar equations are needed. The conditions:

$$y(x_0) = Y_0$$

and:

$$y'(x_0) = Y_1$$

provide those two equations.

More generally, an nth-order equation normally has n arbitrary constants and requires n initial conditions:

$$y(x_0), \quad y'(x_0), \quad \dots, \quad y^{(n-1)}(x_0).$$

The highest derivative prescribed in a standard nth-order IVP is order $n - 1$, not order n. The differential equation itself determines the next derivative once the lower-order data are known.

The Four-Stage Workflow

Stage 1: Solve the differential equation. Find the complete general solution. For a nonhomogeneous linear equation, this means:

$$y = y_h + y_p.$$

Stage 2: Calculate the required derivatives. If the problem supplies $y'(x_0)$, differentiate the complete solution once. For higher-order problems, continue only as far as the highest prescribed derivative.

Stage 3: Apply one condition at a time. Substitute $x = x_0$ into the appropriate formula and record the resulting equation for the constants.

Stage 4: Solve the constant system and verify. Solve all constant equations together. Substitute the constants back into the complete solution, then check both the differential equation and every initial condition.

A First Homogeneous Example

Solve:

$$y'' - y' - 6y = 0,$$

subject to:

$$y(0) = 5, \qquad y'(0) = 0.$$

Step 1: Find the general solution. The characteristic equation is:

$$r^2 - r - 6 = 0.$$

Factor:

$$(r-3)(r+2)=0.$$

The roots are $r = 3, -2$, so the general solution is:

$$y = c_1e^{3x} + c_2e^{-2x}.$$

Because the differential equation is homogeneous, this homogeneous family is already the complete general solution.

Step 2: Differentiate the complete solution. Differentiate each exponential term:

$$y' = 3c_1e^{3x} - 2c_2e^{-2x}.$$

Step 3: Apply $y(0) = 5$. Return to the general solution:

$$y = c_1e^{3x} + c_2e^{-2x}.$$

Substitute $x = 0$ and $y(0) = 5$:

$$5 = c_1e^0 + c_2e^0.$$

Since $e^0 = 1$:

$$c_1 + c_2 = 5.$$

Step 4: Apply $y'(0) = 0$. Return to the derivative:

$$y' = 3c_1 e^{3x} - 2c_2 e^{-2x}.$$

Substitute $x = 0$ and $y'(0) = 0$:

$$0 = 3c_1 e^0 - 2c_2 e^0.$$

Thus:

$$3c_1 - 2c_2 = 0.$$

Step 5: Solve the two equations together. The constant system is:

$$\begin{aligned} c_1 + c_2 &= 5, \\ 3c_1 - 2c_2 &= 0. \end{aligned}$$

Multiply the first equation by 2:

$$2c_1 + 2c_2 = 10.$$

Add this equation to $3c_1 - 2c_2 = 0$. The c_2 terms cancel:

$$5c_1 = 10.$$

Divide by 5:

$$c_1 = 2.$$

Substitute $c_1 = 2$ into $c_1 + c_2 = 5$:

$$2 + c_2 = 5.$$

Subtract 2 from both sides:

$$c_2 = 3.$$

Step 6: Write the selected solution. Substitute both constants into the general solution:

$$\boxed{y = 2e^{3x} + 3e^{-2x}}.$$

Step 7: Check the initial data. At $x = 0$:

$$y(0) = 2 + 3 = 5.$$

The derivative is:

$$y' = 6e^{3x} - 6e^{-2x}.$$

Therefore:

$$y'(0) = 6 - 6 = 0.$$

Both conditions are satisfied.

Block 1 Summary

An IVP is not a new solution method for the differential equation. It is a selection problem performed after the general solution is known.

For a second-order problem:

1. find the complete two-constant solution
2. differentiate it once
3. form one constant equation from each initial condition
4. solve the two equations simultaneously
5. verify both conditions

Apply Conditions To The Complete Solution

The Goal Of This Block

The goal is to prevent the central error in nonhomogeneous IVPs: applying the initial conditions only to y_h while omitting the contribution from y_p.

The Complete Solution Is $y_h + y_p$

For a nonhomogeneous equation:

$$L[y] = g(x),$$

the general solution is:

$$\boxed{y = y_h + y_p}.$$

Although the arbitrary constants appear in y_h, the initial conditions are conditions on the actual solution y. Therefore:

$$y(x_0) = y_h(x_0) + y_p(x_0)$$

and:

$$y'(x_0) = y_h'(x_0) + y_p'(x_0).$$

The particular solution can change the right sides of the equations used to determine the homogeneous constants.

A Complete Forced Example

Solve:

$$y'' + 3y' + 2y = 6x + 7,$$

subject to:

$$y(0) = 2, \qquad y'(0) = -1.$$

Step 1: Find the homogeneous solution. The associated homogeneous characteristic equation is:

$$r^2 + 3r + 2 = 0.$$

Factor:

$$(r + 1)(r + 2) = 0.$$

The roots are $-1, -2$, so:

$$y_h = c_1 e^{-x} + c_2 e^{-2x}.$$

Step 2: Find one particular solution. The forcing $6x + 7$ is linear, so use the complete linear trial:

$$y_p = Ax + B.$$

Differentiate:

$$y_p' = A, \qquad y_p'' = 0.$$

Substitute these expressions into the left side:

$$\begin{aligned} y_p'' + 3y_p' + 2y_p &= 0 + 3A + 2(Ax + B) \\ &= 2Ax + (3A + 2B). \end{aligned}$$

Match this with the required forcing:

$$2Ax + (3A + 2B) = 6x + 7.$$

Matching the coefficients of x gives:

$$2A = 6.$$

Divide by 2:

$$A = 3.$$

Matching the constant terms gives:

$$3A + 2B = 7.$$

Substitute $A = 3$:

$$9 + 2B = 7.$$

Subtract 9 from both sides:

$$2B = -2.$$

Divide by 2:

$$B = -1.$$

Thus:

$$y_p = 3x - 1.$$

Step 3: Write the complete general solution. Use $y = y_h + y_p$:

$$\boxed{y = c_1 e^{-x} + c_2 e^{-2x} + 3x - 1}.$$

This complete formula, not y_h by itself, receives the initial conditions.

Step 4: Differentiate the complete solution. Differentiate every term:

$$y' = -c_1 e^{-x} - 2c_2 e^{-2x} + 3.$$

The constant 3 comes from differentiating the particular term $3x - 1$. It must remain in the derivative condition.

Step 5: Apply $y(0) = 2$. Return to the complete solution:

$$y = c_1 e^{-x} + c_2 e^{-2x} + 3x - 1.$$

Substitute $x = 0$ and $y(0) = 2$:

$$2 = c_1 e^0 + c_2 e^0 + 3(0) - 1.$$

Simplify:

$$2 = c_1 + c_2 - 1.$$

Add 1 to both sides:

$$c_1 + c_2 = 3.$$

Step 6: Apply $y'(0) = -1$. Return to:

$$y' = -c_1 e^{-x} - 2c_2 e^{-2x} + 3.$$

Substitute $x = 0$ and $y'(0) = -1$:

$$-1 = -c_1 e^0 - 2c_2 e^0 + 3.$$

Simplify:

$$-1 = -c_1 - 2c_2 + 3.$$

Subtract 3 from both sides:

$$-4 = -c_1 - 2c_2.$$

Multiply both sides by -1:

$$c_1 + 2c_2 = 4.$$

Step 7: Solve the constant system. The two equations are:

$$\begin{aligned} c_1 + c_2 &= 3, \\ c_1 + 2c_2 &= 4. \end{aligned}$$

Subtract the first equation from the second:

$$c_2 = 1.$$

Substitute $c_2 = 1$ into $c_1 + c_2 = 3$:

$$c_1 + 1 = 3.$$

Subtract 1:

$$c_1 = 2.$$

Step 8: Write and verify the selected solution. Substitute $c_1 = 2, c_2 = 1$ into the complete solution:

$$\boxed{y = 2e^{-x} + e^{-2x} + 3x - 1}.$$

At $x = 0$:

$$y(0) = 2 + 1 + 0 - 1 = 2.$$

The derivative is:

$$y' = -2e^{-x} - 2e^{-2x} + 3.$$

Therefore:

$$y'(0) = -2 - 2 + 3 = -1.$$

Both data are satisfied.

What Goes Wrong If y_p Is Omitted?

In the previous problem:

$$y_h = c_1 e^{-x} + c_2 e^{-2x}, \qquad y_p = 3x - 1.$$

Applying $y(0) = 2$ incorrectly to y_h alone would give:

$$c_1 + c_2 = 2.$$

The correct complete solution gives:

$$c_1 + c_2 - 1 = 2,$$

so:

$$c_1 + c_2 = 3.$$

The two constant equations differ because $y_p(0) = -1$.

Similarly, applying $y'(0) = -1$ only to y_h' would omit:

$$y_p'(0) = 3.$$

This changes the second constant equation as well. The constants live in y_h, but the conditions belong to $y_h + y_p$.

Block 2 Summary

For a nonhomogeneous problem, always write locally:

$$y = y_h + y_p$$

and:

$$y' = y_h' + y_p'.$$

Then apply the data. Omitting y_p or one of its derivatives changes the constant system and usually produces a function that fails the initial conditions.

The Constant System And The Wronskian

The Goal Of This Block

The goal is to see the initial conditions as a linear system for the arbitrary constants and to understand why the Wronskian guarantees that the system can be solved.

Build The General Two-Constant System

Suppose the complete solution is:

$$y = c_1 y_1 + c_2 y_2 + y_p.$$

Differentiate:

$$y' = c_1 y_1' + c_2 y_2' + y_p'.$$

Apply:

$$y(x_0) = Y_0.$$

Substitution gives:

$$c_1 y_1(x_0) + c_2 y_2(x_0) + y_p(x_0) = Y_0.$$

Subtract $y_p(x_0)$ from both sides:

$$c_1 y_1(x_0) + c_2 y_2(x_0) = Y_0 - y_p(x_0).$$

Now apply:

$$y'(x_0) = Y_1.$$

Substitution gives:

$$c_1 y_1'(x_0) + c_2 y_2'(x_0) + y_p'(x_0) = Y_1.$$

Subtract $y_p'(x_0)$:

$$c_1 y_1'(x_0) + c_2 y_2'(x_0) = Y_1 - y_p'(x_0).$$

The constant system is therefore:

$$\boxed{\begin{bmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} Y_0 - y_p(x_0) \\ Y_1 - y_p'(x_0) \end{bmatrix}}.$$

Why The Wronskian Appears Again

The determinant of the constant-system matrix is:

$$\begin{aligned} &y_1(x_0)y_2'(x_0) - y_1'(x_0)y_2(x_0) \\ &= W(y_1, y_2)(x_0). \end{aligned}$$

If y_1, y_2 form a fundamental pair, then their Wronskian is nonzero on the working interval. In particular:

$$W(x_0) \neq 0.$$

Therefore the two-by-two constant system has exactly one solution (c_1, c_2).

This is the algebraic reason a valid pair of initial conditions selects one member of the general solution family.

An Initial Point Away From Zero

Solve:

$$y'' + 4y = 4x,$$

subject to:

$$y\left(\frac{\pi}{4}\right) = 1 + \frac{\pi}{4}, \qquad y'\left(\frac{\pi}{4}\right) = -3.$$

Step 1: Find the complete general solution. The homogeneous equation is:

$$y'' + 4y = 0.$$

Its fundamental pair is:

$$\cos(2x), \qquad \sin(2x).$$

For the forcing $4x$, try:

$$y_p = Ax + B.$$

Then:

$$y_p'' = 0.$$

Substitute into $y_p'' + 4y_p = 4x$:

$$4(Ax + B) = 4x.$$

Match coefficients:

$$4A = 4, \qquad 4B = 0.$$

Thus:

$$A = 1, \qquad B = 0,$$

and:

$$y_p = x.$$

The complete general solution is:

$$y = c_1 \cos(2x) + c_2 \sin(2x) + x.$$

Step 2: Differentiate before substituting the initial point. Differentiate every term:

$$y' = -2c_1 \sin(2x) + 2c_2 \cos(2x) + 1.$$

Step 3: Apply the position condition at $x = \pi/4$. At $x = \pi/4$:

$$2x = \frac{\pi}{2}.$$

Use:

$$\cos\left(\frac{\pi}{2}\right) = 0, \qquad \sin\left(\frac{\pi}{2}\right) = 1.$$

Substitute into the complete solution:

$$1 + \frac{\pi}{4} = c_1(0) + c_2(1) + \frac{\pi}{4}.$$

Subtract $\pi/4$ from both sides:

$$c_2 = 1.$$

Step 4: Apply the derivative condition at $x = \pi/4$. Return to:

$$y' = -2c_1 \sin(2x) + 2c_2 \cos(2x) + 1.$$

Substitute $x = \pi/4$:

$$-3 = -2c_1(1) + 2c_2(0) + 1.$$

Simplify:

$$-3 = -2c_1 + 1.$$

Subtract 1:

$$-4 = -2c_1.$$

Divide by -2:

$$c_1 = 2.$$

Step 5: Write the selected solution. Substitute $c_1 = 2, c_2 = 1$:

$$\boxed{y = 2\cos(2x) + \sin(2x) + x}.$$

The exponentials or trigonometric values are not automatically 1 unless the initial point makes them so. This example required the actual values at $x = \pi/4$.

Block 3 Summary

Initial conditions create a linear system for the arbitrary constants. For a second-order IVP, its coefficient determinant is the Wronskian evaluated at the initial point.

Because a fundamental pair has $W(x_0) \neq 0$, the constant system has one solution. At a nonzero initial point, evaluate every exponential, trigonometric function, polynomial, and derivative at the actual value of x_0.

Existence, Uniqueness, And Zero Initial Data

The Goal Of This Block

The goal is to connect the constant calculations to the existence-and-uniqueness theorem and to distinguish zero data for homogeneous and nonhomogeneous equations.

The Linear Existence-And-Uniqueness Statement

Consider the normalized equation:

$$y'' + P(x)y' + Q(x)y = R(x).$$

If P, Q, R are continuous on an interval I containing x_0, then for any numbers Y_0, Y_1, the IVP:

$$y(x_0) = Y_0, \qquad y'(x_0) = Y_1$$

has exactly one solution on I.

This statement contains two claims:

- **existence:** at least one solution satisfies the equation and the data
- **uniqueness:** no second, different solution satisfies the same equation and the same data on that interval

The continuity interval matters. If normalization divides by a function that vanishes, the theorem cannot be applied across that singular point.

Homogeneous Equation With Zero Data

Consider:

$$y'' + 9y = 0,$$

with:

$$y(0) = 0, \qquad y'(0) = 0.$$

The general solution is:

$$y = c_1 \cos(3x) + c_2 \sin(3x).$$

Differentiate:

$$y' = -3c_1 \sin(3x) + 3c_2 \cos(3x).$$

Apply $y(0) = 0$:

$$0 = c_1 \cos(0) + c_2 \sin(0) = c_1.$$

Thus:

$$c_1 = 0.$$

Apply $y'(0) = 0$:

$$0 = -3c_1 \sin(0) + 3c_2 \cos(0) = 3c_2.$$

Divide by 3:

$$c_2 = 0.$$

Therefore:

$$\boxed{y = 0}.$$

For a homogeneous linear IVP, zero initial data select the zero solution. By uniqueness, no nonzero solution can have the same zero data at the same point.

Nonhomogeneous Equation With Zero Data

Now consider:

$$y'' + y = 1,$$

with the same zero data:

$$y(0) = 0, \qquad y'(0) = 0.$$

The homogeneous solution is:

$$y_h = c_1 \cos x + c_2 \sin x.$$

A constant particular solution $y_p = A$ gives:

$$y_p'' + y_p = 0 + A = A.$$

To reproduce the forcing 1, choose:

$$A = 1.$$

Therefore the complete solution is:

$$y = c_1 \cos x + c_2 \sin x + 1.$$

Differentiate:

$$y' = -c_1 \sin x + c_2 \cos x.$$

Apply $y(0) = 0$:

$$0 = c_1 + 1.$$

Subtract 1:

$$c_1 = -1.$$

Apply $y'(0) = 0$:

$$0 = c_2.$$

Thus:

$$\boxed{y = 1 - \cos x}.$$

Zero initial data do not force the zero function here because $y = 0$ does not satisfy the nonhomogeneous equation:

$$0'' + 0 = 0 \neq 1.$$

The forcing drives the solution away from zero after the initial point.

The Interval Cannot Be Ignored

Consider an equation that becomes, after normalization:

$$y'' + \frac{1}{x}y' = R(x).$$

The coefficient $1/x$ is not continuous at $x = 0$. An IVP based at $x_0 = 1$ may be studied on $(0, \infty)$, while an IVP based at $x_0 = -1$ may be studied on $(-\infty, 0)$.

The existence-and-uniqueness theorem does not justify extending either solution through $x = 0$ without additional analysis.

Block 4 Summary

Continuous normalized coefficients guarantee one solution for each valid pair of initial values on the interval containing x_0.

For a homogeneous equation, zero initial data select $y = 0$. For a nonhomogeneous equation, the same data can select a nonzero forced response. Always check whether the zero function satisfies the differential equation before claiming that zero data imply a zero solution.

Initial Data After Variation Of Parameters

The Goal Of This Block

The goal is to connect Chapter 12's definite-integral construction with the IVP workflow. When the variation integrals begin at the initial point, the chosen particular solution has zero initial value and zero initial derivative there.

Why Anchored Integrals Simplify The Constant Equations

For:

$$y'' + Py' + Qy = R,$$

let y_1, y_2 be a fundamental pair and let W be their Wronskian. Define:

$$u_1(x) = -\int_{x_0}^{x} \frac{y_2(t)R(t)}{W(t)}\,dt$$

and:

$$u_2(x) = \int_{x_0}^{x} \frac{y_1(t)R(t)}{W(t)}\,dt.$$

Then:

$$y_p = u_1 y_1 + u_2 y_2.$$

Because both integrals vanish at $x = x_0$:

$$u_1(x_0) = u_2(x_0) = 0.$$

Therefore:

$$y_p(x_0) = 0.$$

Under the variation-of-parameters auxiliary condition:

$$y_p' = u_1 y_1' + u_2 y_2'.$$

Since both parameter functions vanish at x_0:

$$y_p'(x_0) = 0.$$

The initial data then determine only the homogeneous combination:

$$\begin{aligned} c_1 y_1(x_0) + c_2 y_2(x_0) &= Y_0, \\ c_1 y_1'(x_0) + c_2 y_2'(x_0) &= Y_1. \end{aligned}$$

The complete solution still includes y_p. The anchored construction merely makes its two initial contributions zero.

A Variation-Of-Parameters IVP

Solve on $-\pi/2 < x < \pi/2$:

$$y'' + y = 2\tan x,$$

subject to:

$$y(0) = 1, \qquad y'(0) = 0.$$

Step 1: Find the homogeneous pair and Wronskian. The associated homogeneous equation is:

$$y'' + y = 0.$$

Choose:

$$y_1 = \cos x, \qquad y_2 = \sin x.$$

The homogeneous solution is:

$$y_h = c_1 \cos x + c_2 \sin x.$$

Differentiate the pair:

$$y_1' = -\sin x, \qquad y_2' = \cos x.$$

Compute:

$$\begin{aligned} W &= y_1 y_2' - y_1' y_2 \\ &= \cos^2 x + \sin^2 x \\ &= 1. \end{aligned}$$

Step 2: Construct u_1 from the initial point. The normalized forcing is:

$$R(x) = 2\tan x.$$

Use:

$$u_1(x) = -\int_0^x \frac{y_2(t)R(t)}{W(t)}\,dt.$$

Substitute $y_2(t) = \sin t$, $R(t) = 2\tan t$, and $W = 1$:

$$u_1(x) = -2\int_0^x \sin t \tan t\,dt.$$

Rewrite the integrand:

$$\begin{aligned}\sin t \tan t &= \frac{\sin^2 t}{\cos t}\\ &= \frac{1-\cos^2 t}{\cos t}\\ &= \sec t - \cos t.\end{aligned}$$

Therefore:

$$u_1(x) = -2\int_0^x (\sec t - \cos t)\,dt.$$

On the chosen interval:

$$\int \sec t\,dt = \ln(\sec t + \tan t), \qquad \int \cos t\,dt = \sin t.$$

Apply the limits:

$$\begin{aligned} u_1(x) &= -2[\ln(\sec t + \tan t) - \sin t]_0^x \\ &= -2\ln(\sec x + \tan x) + 2\sin x. \end{aligned}$$

At $x = 0$, both $\ln(1)$ and $\sin(0)$ are zero, as required.

Step 3: Construct u_2 from the initial point. Use:

$$u_2(x) = \int_0^x \frac{y_1(t)R(t)}{W(t)}\, dt.$$

Substitute $y_1(t) = \cos t$, $R(t) = 2\tan t$, and $W = 1$:

$$\begin{aligned} u_2(x) &= \int_0^x 2\cos t \tan t\, dt \\ &= 2\int_0^x \sin t\, dt. \end{aligned}$$

Integrate and apply the limits:

$$\begin{aligned} u_2(x) &= [-2\cos t]_0^x \\ &= -2\cos x + 2 \\ &= 2(1 - \cos x). \end{aligned}$$

Step 4: Reconstruct and simplify y_p. Return to:

$$y_p = u_1 y_1 + u_2 y_2.$$

Substitute the four functions:

$$\begin{aligned} y_p = {} & [-2\ln(\sec x + \tan x) + 2\sin x]\cos x \\ & + 2(1 - \cos x)\sin x. \end{aligned}$$

Expand the second line:

$$\begin{aligned} y_p &= -2\cos x \ln(\sec x + \tan x) + 2\sin x \cos x \\ &\quad + 2\sin x - 2\sin x \cos x. \end{aligned}$$

Cancel the opposite product terms:

$$\boxed{y_p = 2\sin x - 2\cos x \ln(\sec x + \tan x)}.$$

The anchored construction guarantees:

$$y_p(0) = 0, \qquad y_p'(0) = 0.$$

Step 5: Apply the initial data to the complete solution. Write:

$$y = c_1 \cos x + c_2 \sin x + y_p.$$

Apply $y(0) = 1$ using $y_p(0) = 0$:

$$\begin{aligned} 1 &= c_1 \cos(0) + c_2 \sin(0) + 0 \\ &= c_1. \end{aligned}$$

Thus:

$$c_1 = 1.$$

Differentiate the complete solution in structural form:

$$y' = -c_1 \sin x + c_2 \cos x + y_p'.$$

Apply $y'(0) = 0$ using $y_p'(0) = 0$:

$$\begin{aligned} 0 &= -c_1 \sin(0) + c_2 \cos(0) + 0 \\ &= c_2. \end{aligned}$$

Therefore:

$$c_2 = 0.$$

Step 6: Write and verify the selected solution. Substitute the constants and the particular solution:

$$\boxed{y = \cos x + 2 \sin x - 2 \cos x \ln(\sec x + \tan x)}.$$

For a direct equation check, let:

$$h(x) = \ln(\sec x + \tan x).$$

Then:

$$h'(x) = \sec x.$$

The terms $\cos x$ and $2 \sin x$ are homogeneous, so only $z = -2 \cos x\, h$ contributes to $y'' + y$. Differentiate z:

$$\begin{aligned} z' &= 2 \sin x\, h - 2 \cos x\, h' \\ &= 2 \sin x\, h - 2. \end{aligned}$$

Differentiate again:

$$z'' = 2\cos x\, h + 2\sin x \sec x.$$

Add $z = -2\cos x\, h$:

$$\begin{aligned} z'' + z &= 2\sin x \sec x \\ &= 2\tan x. \end{aligned}$$

The differential equation is satisfied.

Block 5 Summary

Variation integrals based at the initial point produce a particular solution with:

$$y_p(x_0) = 0, \qquad y_p'(x_0) = 0.$$

This often makes the constant equations look homogeneous, but the final solution must still include the nonzero forced response $y_p(x)$.

Higher-Order Initial-Value Problems

The Goal Of This Block

The goal is to extend the same workflow to an nth-order equation: calculate the complete n-constant family, differentiate through order $n-1$, and use the n initial conditions to determine all constants.

The General Higher-Order Pattern

An nth-order linear IVP has a differential equation of the form:

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_0(x)y = g(x)$$

and initial data:

$$\begin{aligned} y(x_0) &= Y_0, \\ y'(x_0) &= Y_1, \\ &\vdots \\ y^{(n-1)}(x_0) &= Y_{n-1}. \end{aligned}$$

The complete general solution contains n arbitrary constants. Each condition produces one equation for those constants.

The resulting coefficient matrix is the n-function Wronskian matrix evaluated at x_0. A fundamental homogeneous set makes that determinant nonzero, so the constant system has a unique solution.

A Third-Order Initial-Value Problem

Solve:

$$y''' = 6x,$$

subject to:

$$y(1) = \frac{9}{4}, \qquad y'(1) = 5, \qquad y''(1) = 9.$$

Step 1: Find the complete general solution. Integrate the differential equation once:

$$y'' = 3x^2 + C_1.$$

Integrate again:

$$y' = x^3 + C_1 x + C_2.$$

Integrate a third time:

$$y = \frac{x^4}{4} + \frac{C_1}{2}x^2 + C_2 x + C_3.$$

Rename the arbitrary constants so the complete solution is written more simply:

$$\boxed{y = c_1 + c_2 x + c_3 x^2 + \frac{x^4}{4}}.$$

Step 2: Calculate the required derivatives. Differentiate the renamed formula:

$$y' = c_2 + 2c_3 x + x^3.$$

Differentiate again:

$$y'' = 2c_3 + 3x^2.$$

Step 3: Apply $y''(1) = 9$ first. The highest-derivative condition contains only c_3, so it is the shortest equation to use first.

Substitute $x = 1$ into $y'' = 2c_3 + 3x^2$:

$$9 = 2c_3 + 3.$$

Subtract 3:

$$6 = 2c_3.$$

Divide by 2:

$$c_3 = 3.$$

Step 4: Apply $y'(1) = 5$. Return to:

$$y' = c_2 + 2c_3x + x^3.$$

Substitute $x = 1$, $y'(1) = 5$, and $c_3 = 3$:

$$5 = c_2 + 2(3)(1) + (1)^3.$$

Simplify the numerical terms:

$$5 = c_2 + 6 + 1.$$

Therefore:

$$5 = c_2 + 7.$$

Subtract 7:

$$c_2 = -2.$$

Step 5: Apply $y(1) = 9/4$. Return to the complete solution:

$$y = c_1 + c_2 x + c_3 x^2 + \frac{x^4}{4}.$$

Substitute $x = 1$, $c_2 = -2$, and $c_3 = 3$:

$$\frac{9}{4} = c_1 - 2(1) + 3(1)^2 + \frac{(1)^4}{4}.$$

Simplify:

$$\frac{9}{4} = c_1 - 2 + 3 + \frac{1}{4}.$$

Combine the numerical terms on the right:

$$\frac{9}{4} = c_1 + \frac{5}{4}.$$

Subtract $5/4$:

$$c_1 = 1.$$

Step 6: Write and verify the selected solution. Substitute all three constants:

$$\boxed{y = 1 - 2x + 3x^2 + \frac{x^4}{4}}.$$

Differentiate:

$$y' = -2 + 6x + x^3, \qquad y'' = 6 + 3x^2, \qquad y''' = 6x.$$

At $x = 1$:

$$\begin{aligned} y(1) &= 1 - 2 + 3 + \frac{1}{4} = \frac{9}{4}, \\ y'(1) &= -2 + 6 + 1 = 5, \\ y''(1) &= 6 + 3 = 9. \end{aligned}$$

The equation and all three initial conditions are satisfied.

Using The Easiest Condition First

Initial conditions may be processed in any order. In the third-order example, $y''(1) = 9$ isolated c_3 immediately, then $y'(1) = 5$ isolated c_2, and the remaining condition found c_1.

This triangular structure is common when a differential equation has been solved by repeated integration. For exponential or trigonometric bases, the constants may remain coupled and must be solved simultaneously instead.

Choosing an efficient order is helpful, but every condition must still be applied to the correct derivative of the complete solution.

Block 6 Summary

An nth-order IVP normally needs n initial conditions through derivative order $n-1$. Write all required derivatives before substitution and solve the resulting n equations for the n constants.

When the system is triangular, use the condition that isolates a constant first. When it is coupled, solve all constant equations together.

Verification, Boundaries, And Common Mistakes

The Complete IVP Workflow

Step 1: Identify the order and the data. Count the arbitrary constants expected and confirm that the problem supplies the corresponding number of initial conditions.

Step 2: Choose a valid interval. Normalize the equation if needed and identify singular points. The initial point must lie inside the working interval.

Step 3: Find the complete general solution. For a nonhomogeneous problem, include both parts:

$$y = y_h + y_p.$$

Step 4: Differentiate the complete formula. Calculate every derivative appearing in the data. Carry all particular terms through the differentiation.

Step 5: Apply one condition at a time. Restate the relevant formula, substitute the actual initial point, and write the resulting constant equation.

Step 6: Solve the full constant system. Use substitution, elimination, or matrix methods. Do not assign a constant from one equation while ignoring its coupling to the others.

Step 7: Write the selected solution. Replace every arbitrary constant. The final answer should contain no undetermined c_i.

Step 8: Perform two kinds of verification. Check:

1. the differential equation on the interval
2. every initial condition at x_0

Initial Conditions Versus Boundary Conditions

Initial conditions are imposed at one point:

$$y(x_0) = Y_0, \qquad y'(x_0) = Y_1.$$

Boundary conditions may be imposed at different points, for example:

$$y(a) = A, \qquad y(b) = B.$$

The linear IVP uniqueness theorem does not automatically give the same result for a boundary-value problem. Depending on the equation and boundary data, a boundary-value problem may have:

- exactly one solution
- no solution
- more than one solution

The algebra may still produce a constant system, but its determinant must be examined separately.

Common Mistake: Applying Data To y_h Alone

For a forced problem:

$$y = y_h + y_p.$$

The conditions apply to y, not merely to the portion containing arbitrary constants. Write $y_h + y_p$ locally before substituting the initial point.

Common Mistake: Forgetting A Particular Derivative

If:

$$y = c_1 y_1 + c_2 y_2 + y_p,$$

then:

$$y' = c_1 y_1' + c_2 y_2' + y_p'.$$

The term y_p' can be zero, constant, or a nontrivial function. Calculate it; do not assume it vanishes.

Common Mistake: Treating Every Initial Point As Zero

At $x_0 = 0$:

$$e^{rx_0} = 1.$$

At a nonzero initial point:

$$e^{rx_0}$$

must be evaluated at its actual exponent. The same applies to sine, cosine, polynomial terms, logarithms, and particular solutions.

Common Mistake: Solving Conditions In Isolation

The equations produced by two initial conditions normally share both c_1 and c_2. They form one simultaneous system.

For example:

$$\begin{aligned} c_1 + c_2 &= 5, \\ 2c_1 - c_2 &= 1 \end{aligned}$$

does not allow either constant to be read from one equation alone. Eliminate or substitute using both equations.

Common Mistake: Verifying Only The Data

A function can pass through the required initial values without satisfying the differential equation. Conversely, a member of the general solution family can satisfy the equation but fail the selected data.

A valid IVP solution must pass both tests:

$$\text{differential equation} \quad \text{and} \quad \text{all initial conditions.}$$

Common Mistake: Assuming Zero Data Always Give Zero

Zero data select the zero solution for a homogeneous linear equation. For a forced equation, first test whether $y = 0$ satisfies the equation. If the forcing is nonzero, it does not.

Block 7 Summary

The safest order is:

1. solve the equation
2. write the complete family
3. differentiate completely
4. evaluate at the actual initial point
5. solve all constant equations together
6. verify the equation and the data separately

Initial-value and boundary-value problems may look similar, but their uniqueness behavior is not interchangeable.

Original Practice Set

Practice Problems

Problem 1: Count The Required Initial Data. For a fourth-order linear differential equation:

1. how many arbitrary constants should the general solution contain?
2. what standard initial values would be prescribed at $x = x_0$?

Problem 2: Form And Solve The Constant System. The complete general solution of an equation is:

$$y = c_1 e^x + c_2 e^{-x} + x^2.$$

Find c_1, c_2 if:

$$y(0) = 3, \qquad y'(0) = 0.$$

Problem 3: Use A Nonzero Initial Point. The complete general solution is:

$$y = c_1 \cos x + c_2 \sin x + 2.$$

Find c_1, c_2 if:

$$y\left(\frac{\pi}{2}\right) = 5, \qquad y'\left(\frac{\pi}{2}\right) = -2.$$

Problem 4: Diagnose Zero Initial Data. Explain why zero initial data do not imply $y = 0$ for:

$$y'' - 2y' + y = e^x, \qquad y(0) = 0, \qquad y'(0) = 0.$$

Problem 5: Solve A Homogeneous IVP. Solve:

$$y'' + 2y' - 3y = 0,$$

subject to:

$$y(0) = 2, \qquad y'(0) = -2.$$

Problem 6: Solve A Repeated-Root IVP. Solve:

$$y'' - 4y' + 4y = 0,$$

subject to:

$$y(0) = 1, \qquad y'(0) = 0.$$

Problem 7: Include A Polynomial Particular Solution. Solve:

$$y'' + 4y = 8x,$$

subject to:

$$y(0) = 2, \qquad y'(0) = 1.$$

Problem 8: Include An Exponential Particular Solution. Solve:

$$y'' - 3y' + 2y = 6e^{-x},$$

subject to:

$$y(0) = 0, \qquad y'(0) = 1.$$

Problem 9: Apply Trigonometric Data Away From Zero. Solve:

$$y'' + 9y = 0,$$

subject to:

$$y\left(\frac{\pi}{6}\right) = 2, \qquad y'\left(\frac{\pi}{6}\right) = 3.$$

Problem 10: Use An Anchored Particular Solution. For the equation:

$$y'' + y = 3\tan x, \qquad -\frac{\pi}{2} < x < \frac{\pi}{2},$$

variation of parameters with lower limit 0 produces:

$$y_p = 3\sin x - 3\cos x \ln(\sec x + \tan x).$$

Use this particular solution to solve the IVP:

$$y(0) = -2, \qquad y'(0) = 4.$$

Problem 11: Solve A Third-Order IVP. Solve:

$$y''' = 12x,$$

subject to:

$$y(0) = 1, \qquad y'(0) = -1, \qquad y''(0) = 2.$$

Problem 12: Test Candidate Solutions Completely. For the IVP:

$$y'' - y = 0, \qquad y(0) = 2, \qquad y'(0) = 0,$$

determine which candidates are valid. For each rejected candidate, state whether it fails the differential equation or an initial condition.

(a)

$$y = 2\cosh x$$

(b)

$$y = 2e^x$$

(c)

$$y = 2$$

Worked Answers: Problems 1 To 4

Answer 1. A fourth-order linear equation has a general solution containing four arbitrary constants.

The standard initial data at $x = x_0$ are:

$$\boxed{y(x_0) = Y_0, \quad y'(x_0) = Y_1, \quad y''(x_0) = Y_2, \quad y'''(x_0) = Y_3}.$$

The highest prescribed derivative is third order, one order below the differential equation.

Answer 2. The complete solution is:

$$y = c_1 e^x + c_2 e^{-x} + x^2.$$

Differentiate every term:

$$y' = c_1 e^x - c_2 e^{-x} + 2x.$$

Apply $y(0) = 3$:

$$\begin{aligned} 3 &= c_1 e^0 + c_2 e^0 + (0)^2 \\ &= c_1 + c_2. \end{aligned}$$

Thus:

$$c_1 + c_2 = 3.$$

Apply $y'(0) = 0$:

$$\begin{aligned} 0 &= c_1 e^0 - c_2 e^0 + 2(0) \\ &= c_1 - c_2. \end{aligned}$$

Thus:

$$c_1 - c_2 = 0.$$

The second equation gives:

$$c_1 = c_2.$$

Substitute $c_2 = c_1$ into $c_1 + c_2 = 3$:

$$2c_1 = 3.$$

Divide by 2:

$$c_1 = \frac{3}{2}.$$

Therefore:

$$c_2 = \frac{3}{2}.$$

The selected solution is:

$$\boxed{y = \frac{3}{2}e^{x} + \frac{3}{2}e^{-x} + x^2}.$$

Answer 3. The complete solution is:

$$y = c_1 \cos x + c_2 \sin x + 2.$$

Differentiate:

$$y' = -c_1 \sin x + c_2 \cos x.$$

At $x = \pi/2$:

$$\cos\left(\frac{\pi}{2}\right) = 0, \qquad \sin\left(\frac{\pi}{2}\right) = 1.$$

Apply $y(\pi/2) = 5$:

$$5 = c_1(0) + c_2(1) + 2.$$

Subtract 2:

$$c_2 = 3.$$

Apply $y'(\pi/2) = -2$:

$$-2 = -c_1(1) + c_2(0).$$

Therefore:

$$c_1 = 2.$$

The selected solution is:

$$\boxed{y = 2\cos x + 3\sin x + 2}.$$

Answer 4. The IVP is:

$$y'' - 2y' + y = e^x, \qquad y(0) = 0, \qquad y'(0) = 0.$$

Test the zero function in the differential equation. For $y = 0$:

$$y' = 0, \qquad y'' = 0.$$

Therefore the left side is:

$$y'' - 2y' + y = 0 - 0 + 0 = 0.$$

The required right side is:

$$e^x,$$

which is not identically zero. Hence $y = 0$ does not satisfy the differential equation, even though it satisfies the two initial conditions.

The nonzero forcing creates a nonzero response after the initial point. Zero initial data imply a zero solution only for a homogeneous linear equation.

Worked Answers: Problems 5 To 8

Answer 5. Solve:

$$y'' + 2y' - 3y = 0,$$

with:

$$y(0) = 2, \qquad y'(0) = -2.$$

The characteristic equation is:

$$r^2 + 2r - 3 = 0.$$

Factor:

$$(r-1)(r+3)=0.$$

The roots are $1, -3$, so:

$$y=c_1e^x+c_2e^{-3x}.$$

Differentiate:

$$y'=c_1e^x-3c_2e^{-3x}.$$

Apply $y(0)=2$:

$$c_1+c_2=2.$$

Apply $y'(0)=-2$:

$$c_1-3c_2=-2.$$

Subtract the first equation from the second:

$$-4c_2=-4.$$

Divide by -4:

$$c_2=1.$$

Substitute into $c_1 + c_2 = 2$:

$$c_1 + 1 = 2.$$

Thus:

$$c_1 = 1.$$

Therefore:

$$\boxed{y = e^x + e^{-3x}}.$$

Answer 6. Solve:

$$y'' - 4y' + 4y = 0,$$

with:

$$y(0) = 1, \qquad y'(0) = 0.$$

The characteristic equation is:

$$r^2 - 4r + 4 = 0.$$

Factor:

$$(r - 2)^2 = 0.$$

The repeated root is $r = 2$, so:

$$y = (c_1 + c_2 x)e^{2x}.$$

Differentiate with the product rule:

$$\begin{aligned} y' &= c_2 e^{2x} + 2(c_1 + c_2 x)e^{2x} \\ &= (c_2 + 2c_1 + 2c_2 x)e^{2x}. \end{aligned}$$

Apply $y(0) = 1$:

$$1 = (c_1 + c_2(0))e^0 = c_1.$$

Thus:

$$c_1 = 1.$$

Apply $y'(0) = 0$:

$$0 = (c_2 + 2c_1 + 2c_2(0))e^0.$$

Substitute $c_1 = 1$:

$$0 = c_2 + 2.$$

Therefore:

$$c_2 = -2.$$

The selected solution is:

$$\boxed{y = (1 - 2x)e^{2x}}.$$

Answer 7. Solve:

$$y'' + 4y = 8x,$$

with:

$$y(0) = 2, \qquad y'(0) = 1.$$

The homogeneous solution is:

$$y_h = c_1 \cos(2x) + c_2 \sin(2x).$$

For the polynomial forcing, try:

$$y_p = Ax + B.$$

Then:

$$y_p'' = 0.$$

Substitute into $y_p'' + 4y_p = 8x$:

$$4(Ax + B) = 8x.$$

Match coefficients:

$$A = 2, \qquad B = 0.$$

Thus:

$$y_p = 2x.$$

The complete solution is:

$$y = c_1 \cos(2x) + c_2 \sin(2x) + 2x.$$

Differentiate:

$$y' = -2c_1 \sin(2x) + 2c_2 \cos(2x) + 2.$$

Apply $y(0) = 2$:

$$2 = c_1.$$

Apply $y'(0) = 1$:

$$1 = 2c_2 + 2.$$

Subtract 2:

$$-1 = 2c_2.$$

Divide by 2:

$$c_2 = -\frac{1}{2}.$$

Therefore:

$$\boxed{y = 2\cos(2x) - \frac{1}{2}\sin(2x) + 2x}.$$

Answer 8. Solve:

$$y'' - 3y' + 2y = 6e^{-x},$$

with:

$$y(0) = 0, \qquad y'(0) = 1.$$

The associated characteristic equation is:

$$r^2 - 3r + 2 = 0.$$

Factor:

$$(r-1)(r-2) = 0.$$

Therefore:

$$y_h = c_1 e^x + c_2 e^{2x}.$$

For the forcing $6e^{-x}$, try:

$$y_p = Ae^{-x}.$$

Differentiate:

$$y_p' = -Ae^{-x}, \qquad y_p'' = Ae^{-x}.$$

Substitute into the left side:

$$\begin{aligned} y_p'' - 3y_p' + 2y_p &= Ae^{-x} - 3(-Ae^{-x}) + 2Ae^{-x} \\ &= 6Ae^{-x}. \end{aligned}$$

Match with $6e^{-x}$:

$$6A = 6.$$

Thus:

$$A = 1, \qquad y_p = e^{-x}.$$

The complete solution is:

$$y = c_1e^x + c_2e^{2x} + e^{-x}.$$

Differentiate:

$$y' = c_1e^x + 2c_2e^{2x} - e^{-x}.$$

Apply $y(0) = 0$:

$$0 = c_1 + c_2 + 1.$$

Subtract 1:

$$c_1 + c_2 = -1.$$

Apply $y'(0) = 1$:

$$1 = c_1 + 2c_2 - 1.$$

Add 1:

$$c_1 + 2c_2 = 2.$$

Subtract the first constant equation from the second:

$$c_2 = 3.$$

Substitute into $c_1 + c_2 = -1$:

$$c_1 + 3 = -1.$$

Therefore:

$$c_1 = -4.$$

The selected solution is:

$$\boxed{y = -4e^{x} + 3e^{2x} + e^{-x}}.$$

Worked Answers: Problems 9 To 12

Answer 9. Solve:

$$y'' + 9y = 0,$$

with:

$$y\left(\frac{\pi}{6}\right) = 2, \qquad y'\left(\frac{\pi}{6}\right) = 3.$$

The characteristic roots are $\pm 3i$, so:

$$y = c_1 \cos(3x) + c_2 \sin(3x).$$

Differentiate:

$$y' = -3c_1 \sin(3x) + 3c_2 \cos(3x).$$

At $x = \pi/6$:

$$3x = \frac{\pi}{2}.$$

Therefore:

$$\cos(3x) = 0, \qquad \sin(3x) = 1.$$

Apply the first condition:

$$2 = c_1(0) + c_2(1).$$

Thus:

$$c_2 = 2.$$

Apply the derivative condition:

$$3 = -3c_1(1) + 3c_2(0).$$

Divide by -3:

$$c_1 = -1.$$

Hence:

$$\boxed{y = -\cos(3x) + 2\sin(3x)}.$$

Answer 10. The equation and anchored particular solution are:

$$y'' + y = 3\tan x$$

and:

$$y_p = 3\sin x - 3\cos x \ln(\sec x + \tan x).$$

The homogeneous solution is:

$$y_h = c_1 \cos x + c_2 \sin x.$$

Thus the complete solution is:

$$y = c_1 \cos x + c_2 \sin x + y_p.$$

Because the variation integrals were based at 0:

$$y_p(0) = 0, \qquad y_p'(0) = 0.$$

Apply $y(0) = -2$:

$$\begin{aligned} -2 &= c_1 \cos(0) + c_2 \sin(0) + y_p(0) \\ &= c_1. \end{aligned}$$

Therefore:

$$c_1 = -2.$$

Differentiate the complete solution structurally:

$$y' = -c_1 \sin x + c_2 \cos x + y_p'.$$

Apply $y'(0) = 4$:

$$\begin{aligned} 4 &= -c_1 \sin(0) + c_2 \cos(0) + y_p'(0) \\ &= c_2. \end{aligned}$$

Thus:

$$c_2 = 4.$$

Substitute the constants and y_p:

$$\boxed{y = -2\cos x + 4\sin x + 3\sin x - 3\cos x \ln(\sec x + \tan x)}.$$

Combine the two sine terms:

$$\boxed{y = -2\cos x + 7\sin x - 3\cos x \ln(\sec x + \tan x)}.$$

Answer 11. Solve:

$$y''' = 12x,$$

with:

$$y(0) = 1, \qquad y'(0) = -1, \qquad y''(0) = 2.$$

Integrate the differential equation once:

$$y'' = 6x^2 + C_1.$$

Integrate again:

$$y' = 2x^3 + C_1 x + C_2.$$

Integrate a third time:

$$y = \frac{x^4}{2} + \frac{C_1 x^2}{2} + C_2 x + C_3.$$

Rename the constants:

$$y = c_1 + c_2 x + c_3 x^2 + \frac{x^4}{2}.$$

Then:

$$y' = c_2 + 2c_3 x + 2x^3$$

and:

$$y'' = 2c_3 + 6x^2.$$

Apply $y(0) = 1$:

$$c_1 = 1.$$

Apply $y'(0) = -1$:

$$c_2 = -1.$$

Apply $y''(0) = 2$:

$$2c_3 = 2.$$

Divide by 2:

$$c_3 = 1.$$

Therefore:

$$\boxed{y = 1 - x + x^2 + \frac{x^4}{2}}.$$

Answer 12. The IVP is:

$$y'' - y = 0, \qquad y(0) = 2, \qquad y'(0) = 0.$$

Candidate (a): $y = 2\cosh x$**.** Differentiate:

$$y' = 2\sinh x, \qquad y'' = 2\cosh x.$$

Check the differential equation:

$$y'' - y = 2\cosh x - 2\cosh x = 0.$$

Check the data:

$$y(0) = 2\cosh(0) = 2,$$

and:

$$y'(0) = 2\sinh(0) = 0.$$

Candidate (a) is valid.

Candidate (b): $y = 2e^x$. Since $y'' = 2e^x$, the differential equation is satisfied:

$$y'' - y = 0.$$

Also:

$$y(0) = 2.$$

However:

$$y' = 2e^x,$$

so:

$$y'(0) = 2 \neq 0.$$

Candidate (b) is rejected because it fails the derivative condition.

Candidate (c): $y = 2$. The initial values are:

$$y(0) = 2, \qquad y'(0) = 0.$$

However:

$$y'' = 0.$$

Therefore:

$$y'' - y = 0 - 2 = -2 \neq 0.$$

Candidate (c) is rejected because it fails the differential equation.

Chapter 13 Final Summary

For a second-order nonhomogeneous equation, begin with the complete solution:

$$\boxed{y = c_1 y_1 + c_2 y_2 + y_p}.$$

Differentiate the complete formula:

$$\boxed{y' = c_1 y_1' + c_2 y_2' + y_p'}.$$

At the initial point x_0, the data produce:

$$\begin{aligned} c_1 y_1(x_0) + c_2 y_2(x_0) &= Y_0 - y_p(x_0), \\ c_1 y_1'(x_0) + c_2 y_2'(x_0) &= Y_1 - y_p'(x_0). \end{aligned}$$

The coefficient determinant is the Wronskian $W(x_0)$. A fundamental pair has $W(x_0) \neq 0$, so the constants are uniquely determined.

For an nth-order equation, use n initial conditions through derivative order $n-1$. After selecting the constants, verify two separate requirements:

$$\boxed{\text{the differential equation} \quad \text{and} \quad \text{every initial condition}}.$$

The chapter's final lesson is:

> **initial data select from the complete solution family, not from the homogeneous part alone; differentiate every term, evaluate at the actual initial point, solve the full constant system, and verify both the equation and the data.**